

Cloud computing Lab 2

Name : Lakshay Mittal

SRN : PES2UG23CS300

Ss1:

 **Fest Monolith**
FastAPI • SQLite • Locust

Logged in as PES2UG23CS300

Events

My Events

Checkout

Logout

 **Events**

View My Events →

Welcome PES2UG23CS300. Register for events below.

Event ID: 1

₹ 500

Hackathon

Includes certificate • instant registration • limited seats

Register

Event ID: 2

₹ 300

Dance

Includes certificate • instant registration • limited seats

Register

Event ID: 3

₹ 500

Hackathon

Includes certificate • instant registration • limited seats

Register

Event ID: 4

₹ 300

Dance Battle

Includes certificate • instant registration • limited seats

Register

Event ID: 5

₹ 400

AI Workshop

Includes certificate • instant registration • limited seats

Register

Event ID: 6

₹ 200

Photography Walk

Includes certificate • instant registration • limited seats

Register

Event ID: 7

₹ 350

Event ID: 8

₹ 250

Event ID: 9

₹ 150

SS2:

 **Fest Monolith**
FastAPI • SQLite • Locust

Login

Create Account

 **Monolith Failure**

HTTP 500

One bug in one module impacted the **entire application**.

Error Message

division by zero

Why did this happen?

Because this is a **monolithic application**: all modules share the same runtime and deployment. When one feature crashes, it affects the whole system.

What should you do in the lab?

- Take a screenshot (crash demonstration)
- Fix the bug in the indicated module
- Restart the server and verify recovery

Back to Events

Login

CC Week X • Monolithic Applications Lab

```
INFO: 127.0.0.1:52331 - "GET /checkout HTTP/1.1" 500 Internal Server Error
ERROR: Exception in ASGI application
```

SS3

 **Fest Monolith**
FastAPI • SQLite • Locust

LoginCreate Account

Checkout

This route is used to demonstrate a monolith crash + optimization.

Total Payable
₹ 6600

✓ After fixing + optimizing checkout logic, re-run Locust and compare results.

What you should observe

- One buggy feature can crash the entire monolith.
- Inefficient loops cause high response times under load.
- Optimization improves performance but architecture still scales as one unit.

Next Lab: Split this monolith into Microservices (Events / Registration / Checkout).

CC Week X • Monolithic Applications Lab

INFO: 127.0.0.1:64908 - "GET /checkout HTTP/1.1" 200 OK

SS4 (checkout before optimization)

 **LOCUST**

Hosthttp://localhost:8000StatusSTOPPEDRPS0.6Failures0%NEWRESET

STATISTICSCHARTSFAILURES EXCEPTIONSCURRENT RATIODOWNLOAD DATALOGS



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	18	0	8	2000	2000	120.85	5	2039	2797	0.6	0
	Aggregated	18	0	8	2000	2000	120.85	5	2039	2797	0.6	0

```

lback
    def python_check_callback(self, watcher_ptr): # pylint:disable
=unused-argument

KeyboardInterrupt
2026-01-30T09:38:46Z
[2026-01-30 15:08:46,487] lakshaymittal/INFO/locust.main: Shutting
down (exit code 0)
Type      Name  # reqs      # fails |    Avg    Min    Max    Med
|  req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /checkout    18      0(0.00%) |    120     5   2039
      8 |    0.61      0.00
-----|-----|-----|-----|-----|-----|
      Aggregated    18      0(0.00%) |    120     5   2039
      8 |    0.61      0.00

Response time percentiles (approximated)
Type      Name      50%    66%    75%    80%    90%    95%    98%
99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /checkout      8      8      9      9      11   2000    2
000   2000   2000   2000   2000    18
-----|-----|-----|-----|-----|-----|-----|
      Aggregated      8      8      9      9      11   2000
2000   2000   2000   2000   2000    18

(.venv) PS C:\Users\dipak\Documents\PES2UG23CS300\Monolith_CC_Lab-
2>

```

ss5

Host

http://localhost:8000

Status

STOPPED

RPS

0.7

Failures

0%

NEW

RESET

⚙️

STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOWNLOAD DATA
LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	19	0	7	2100	2100	115.69	4	2072	2797	0.7	0
	Aggregated	19	0	7	2100	2100	115.69	4	2072	2797	0.7	0

```

2026-01-30T09:43:26Z
[2026-01-30 15:13:26,955] lakshaymittal/INFO/locust.main: Shutting
down (exit code 0)
Type      Name  # reqs      # fails |   Avg   Min    Max   Med
| req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /checkout    19      0(0.00%) |   115    4   2072
7 |    0.63    0.00
-----|-----|-----|-----|-----|-----|
Aggregated    19      0(0.00%) |   115    4   2072
7 |    0.63    0.00

Response time percentiles (approximated)
Type      Name      50%    66%    75%    80%    90%    95%    98%
99%    99.9% 99.99% 100% # reqs

```

Ss6: (events before optimization)

Host

http://localhost:8000

Status

STOPPED

RPS

0.5

Failures

0%

NEW

RESET

⚙️

STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOWNLOAD DATA
LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	16	0	330	2300	2300	452.24	289	2336	21138	0.5	0
	Aggregated	16	0	330	2300	2300	452.24	289	2336	21138	0.5	0

```

KeyboardInterrupt
2026-01-30T09:56:30Z
[2026-01-30 15:26:30,243] lakshaymittal/INFO/locust.main: Shutting down (exit code 0)
Type      Name  # reqs      # fails |      Avg      Min      Max      Me
| req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      16      0(0.00%) |      452
289      2336      330 |      0.53      0.00
-----|-----|-----|-----|-----|-----|-----|-----|
Aggregated      16      0(0.00%) |      452      289      2336
330 |      0.53      0.00

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      330      350      350      350
370      2300      2300      2300      2300      2300      16
-----|-----|-----|-----|-----|-----|-----|-----|
Aggregated      330      350      350      350      370      2300
2300      2300      2300      2300      2300      16

```

Ss7: (events after optimization)



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	18	0	7	2100	2100	122.09	5	2074	21138	0.7	0
Aggregated		18	0	7	2100	2100	122.09	5	2074	21138	0.7	0

```
[2026-01-30 15:37:13,842] lakshaymittal/INFO/locust.main: Shutting
down (exit code 0)
Type      Name      # reqs      # fails |      Avg      Min      Max      Med
|      req/s      failures/s
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      18      0(0.00%) |      122
      4      2074      7 |      0.63      0.00
-----|-----|-----|-----|-----|-----|-----|
      Aggregated      18      0(0.00%) |      122      4      2074
      7 |      0.63      0.00

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      8      8      8      8
      9      2100      2100      2100      2100      2100      18
-----|-----|-----|-----|-----|-----|-----|
      Aggregated      8      8      8      8      9      2100
2100      2100      2100      2100      2100      18
```

ss8

 LOCUST

Host

http://localhost:8000

Status

STOPPED

RPS

0.7

Failures

0%

NEW

RESET



STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	19	0	110	2200	2200	225.04	99	2186	3144	0.7	0
	Aggregated	19	0	110	2200	2200	225.04	99	2186	3144	0.7	0

```
KeyboardInterrupt
2026-01-30T10:09:37Z
[2026-01-30 15:39:37,102] lakshaymittal/INFO/locust.main: Shutting
down (exit code 0)
Type      Name      # reqs      # fails |      Avg      Min      Max      Med
| req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      19      0(0.00%) |      225
      98      2186      110 |      0.64      0.00
-----|-----|-----|-----|-----|-----|-----|
      Aggregated      19      0(0.00%) |      225      98      2186
      110 |      0.64      0.00

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      110      130      130      140
      150      2200      2200      2200      2200      2200      2200      19
-----|-----|-----|-----|-----|-----|-----|
      Aggregated      110      130      130      140      150      2200
2200      2200      2200      2200      2200      19
```

Ss9 (myevents after optimization)

 LOCUST

Host

http://localhost:8000

Status

CLEANUP

RPS

0.6

Failures

0%

EDIT

STOP

RESET



STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	18	0	7	2100	2100	122.28	5	2075	3144	0.6	0
	Aggregated	18	0	7	2100	2100	122.28	5	2075	3144	0.6	0

```

KeyboardInterrupt
2026-01-30T10:11:56Z
[2026-01-30 15:41:56,319] lakshaymittal/INFO/locust.main: Shutting
down (exit code 0)
Type      Name      # reqs      # fails |   Avg   Min   Max   Med
| req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      18      0(0.00%) |   122
      4      2075      7 |   0.61      0.00
-----|-----|-----|-----|-----|-----|
      Aggregated      18      0(0.00%) |   122      4      2075
      7 |   0.61      0.00

```

```

Response time percentiles (approximated)
Type      Name      50%   66%   75%   80%   90%   95%   98%
99% 99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      7      8      8      9
      12      2100      2100      2100      2100      2100      2100      18
-----|-----|-----|-----|-----|-----|
      Aggregated      7      8      8      9      12      2100
2100      2100      2100      2100      2100      18

```

```

(.venv) PS C:\Users\dipak\Documents\PES2UG23CS300\Monolith_CC_Lab-
2>

```

/events route

Bottleneck: The route was slowed down by dummy for loops that added processing time without affecting the output.

Change made: These unnecessary loops were removed from the route logic.

Why it improved: Removing redundant computations reduced CPU usage per request, resulting in nearly a 50% improvement in average response time (350 ms → 108 ms).

/my_events route

Bottleneck: The presence of non-essential iteration logic increased execution time.

Change made: The dummy loops were removed.

Why it improved: Fewer instructions were executed per request, allowing the route to respond faster and reducing average latency from 210 ms to 108 ms.